

SANJAY SUTHRAYE

SENIOR DATA SCIENTIST | AI & ML ENGINEER

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SUMMARY

AI Engineer and Full Stack Data Scientist with **7 years** experience building production artificial intelligence systems throughout the classical-to-generative spectrum, from prescriptive modelling and **Time-Series Forecasting** to **Agentic Frameworks (LangGraph, MCP)**, RAG and GraphRAG. Owns end-to-end solutions: statistical modelling (**XGBoost, LSTM, SARIMAX**), LLM platform design and deployment as containerised microservices (**Docker, FastAPI, AWS**). Demonstrated at **AstraZeneca** and **Tesco** to turn data into scalable, quantifiable business effect, with a record of technical mentorship, conference presentation (**PMSA**) and award-acknowledged execution of enterprise AI solutions.

SKILLS

Generative AI & LLMs	Fine-tuning LLMs, Multi-Agent Systems, Agentic Workflows (LangChain/LangGraph, CrewAI, AutoGen, Strands), Model Context Protocol (MCP) & A2A Protocols, RAG & GraphRAG, Knowledge Graphs (KGs), Vector Databases, Custom Short-Term/Long-Term Memories, Observability, Online/Offline Evaluations, LLMs, NLP (NER, Sentiment Analysis)
Deep Learning	MLPs, DNN, CNN, RNN, LSTM, Auto-encoders, Transformers, Sentence Transformers, MoE (Mixture of Experts) Architecture
Statistics & Machine Learning	Predictive Modelling, Probabilistic Modelling, Regression (Linear/Logistic), Random Forest, XGBoost, LightGBM, CatBoost, Survival Analysis, Clustering (K-Means, DBSCAN), Weak Supervision (Snorkel), Semi-supervised Learning (PU Learning), Active Learning, Model Explainability (SHAP, LIME), Time Series (Univariate, Multivariate; SARIMAX, LSTM), Inferential Statistics, Causal Impact Analysis, A/B Testing, Prospective & Retrospective Experiment Design
Core Competencies	Customer Segmentation, Churn Prediction, Market Mix Modelling, Pricing Models, ROI Optimization
Programming & Tools	Python, R, SQL, PySpark, NoSQL, Distributed SQL, Scikit-learn, Pandas, NumPy, SciPy, spaCy, NLTK, FastAPI, Snowpark, Git, Jira
AWS (3.5 years)	Bedrock, AgentCore, SageMaker, Step Functions, EC2, RDS, EMR, ECS, S3, Lambda, Vector Stores, Build, CodeCommit
Azure & Databricks (4+ years)	MLflow, MLOps, CI/CD, Data Warehouses, Data Lakes, also GCP Vertex AI, Docker, Kubernetes, Apache Airflow, Snowflake, BigQuery, Apache Kafka, Tableau, Power BI
Leadership	Led cross-functional teams aligning AI research, engineering, and business strategy to deliver scalable enterprise AI solutions; stakeholder management, technical delivery, Agile/Scrum, Ethical & Responsible AI

WORK EXPERIENCE

ASTRAZENECA

Domain: Pharmaceuticals/ Healthcare

Team: Oncology Business Unit

AI Engineering Specialist
Data Scientist

Sep'25 - Present
Apr'23 - Sep'25

- Architected **Cami**, an Agentic AI Assistant (**AWS Bedrock, Claude/Gemini**) for Breast Cancer Commercial teams. An orchestration layer routes compliance-screened, decomposed queries in parallel to a **Text-to-SQL agent (Trino)**, a source-traceable retrieval agent, and a **KOL/referral agent**.

Cut analysis turnaround from ~6 weeks to <1 week and replaced \$100K+/year vendor analytics with infrastructure-only cost.

- Developed an **anchor-based feature pipeline (PySpark/EMR, SageMaker)** across ~964M **medical claim records (~900K patients)** to infer cancer stage, progression, and recurrence in **endometrial cancer** – one shared framework answering **four clinically distinct questions** (metastatic-at-diagnosis, 60–120 day progression, recurrence, surgical timing), thousands of features reduced via **mutual-information selection**, **gradient-boosted classifiers** with **Bayesian tuning**, **MLflow**, and **SHAP**, scored across **178K patients**.
- Built **PySpark pipelines on EMR (Step Functions, Terraform)** across **billions of claim records**, re-assembling the clinical journey for each **lung-cancer patient** by combining deterministic claims rules with **ML-derived clinical flags**. Assigned patients to a single clinician/site, predicted to true market size, and district-benchmarked **19K accounts** across **18 opportunity types** for **four medicines** each quarter into actionable field dashboards.
- **Led innovation in the Commercial GenAI** space by fine-tuning **Mistral-7B** using **Supervised Finetuning (SFT-LoRA)** for **HR (Hormone Receptor)** and **HER2 (Human Epidermal Growth Factor Receptor 2)** Biomarker enrichments, ways to summarize patient journey and physician behaviors.
- Role combines **delivery (70%)** and **applied AI research (30%)**, leading a team of **8+ members** across multiple parallel projects spanning solution architecture, model development, and production deployment. Formally an **IC**, but acts as a fully hands-on **Project and Capability Leader** to ensure smooth deliveries.
- Responsible for **solution architecture**, **delivery leadership**, and **stakeholder/vendor management** across AI initiatives.
- Recognized with the '**Annual Award 2024**' in the '**Lead with AI and Insights**' category for work on Patient Medical Journey Summarization using LLMs and fine-tuning **Mistral-7B** for Biomarker Enrichment.

TESCO

Domain: UK Retail Industry

Team: Marketing Performance & Optimization

Senior Applied Data Scientist

Aug'21 - Apr'23

- Developed and containerised a **Churn Prediction** Microservice employing XGBoost and CatBoost, overseeing more than **200k homes**. Validated model effectiveness by rigorous **A/B testing**, yielding a **38% enhancement in retention**.
- Engineered a scalable **Product Choice Optimiser** and **Market Mix Modelling** framework utilising PySpark, managing millions of transactions. Employed **Adstock** transforms to model advertising decay and applied multi-objective optimisation to achieve substantial enhancements in **return on advertising investment**.
- Automated **High Value Customer (HVC) segmentation processes** employing K-Means++ on Hadoop clusters to efficiently categorise customers according to behavioural characteristics.
- Implemented machine learning models to identify clients with significant **growth potential inside cohorts**, enabling dynamic tracking that informed targeted marketing activities.
- **Led a hackathon-winning project** (April 2022) focused on employee wellness, improving workplace welfare.
- Recognised as **Employee of the Month** (April 2022) within the Spotlight program for contributions to data science.

CARTESIAN CONSULTING

Domain: Consulting

Senior Data Analyst

Feb'21 - Aug'21

- Developed a machine learning model utilising the **Levenshtein Algorithm** to identify analogous items, employing the Pareto principle to concentrate on the top 20% of suggestions.
- Developed the **Dealer Health Index** from inception, managing all aspects from data acquisition to visualisation in Tableau and Excel.
- Enhanced metrics for **62% of subpar dealers** using data-driven advice.
- Employed text analysis and **similarity scoring** to enhance product portfolios and improve sales efficiency.

DXC TECHNOLOGY

Domain: IT Services

Senior Analyst at Helpdesk
Technical Analyst (Contractor)

Mar'19 - Apr'20
Jun'18 - Mar'19

- Developed a machine learning model utilising **named-entity recognition** to evaluate **customer re-views** and discern primary sentiment influencers.
- Enhanced team traffic procedures by regression analysis, **reducing average handling time by 15%**.
- Aligned helpdesk operations with industry standards via data-driven workflow enhancements.
- **Administered SQL/MS SQL databases** and upheld a centralised knowledge repository that enhanced information dissemination.
- Examined **call flow data** to discern trends that enhanced first-call resolution rates.

AWARDS AND PRESENTATIONS

- **PMSA Global Summit 2024** – Patient Journey Summarization and Biomarker Enrichment using LLMs (pmsa.org/other-events/summit/2024-summit).
- **AstraZeneca 2023 Annual Award – Lead with AI and Insights** – Patient’s medical journey summarization and Biomarker Enrichment using manually fine-tuned LLMs.
- **Led a hackathon-winning project** at Tesco (February 2022) focused on employee wellness, improving workplace welfare.
- Recognised as **Employee of the Month** at Tesco (May 2022) within the Spotlight program.

EDUCATION

University of Hyderabad
Post Graduate Diploma in **Artificial Intelligence and Machine Learning**

Jan'21 - Jan'22

AMC College of Engineering
Bachelor of Engineering

Jun'14 - Jun'18

LANGUAGES

English - Fluent

German - A2 (Enrolled - Intensive Course)